CGS 2025 Project Report: Growing a Sympoietic Art Organism

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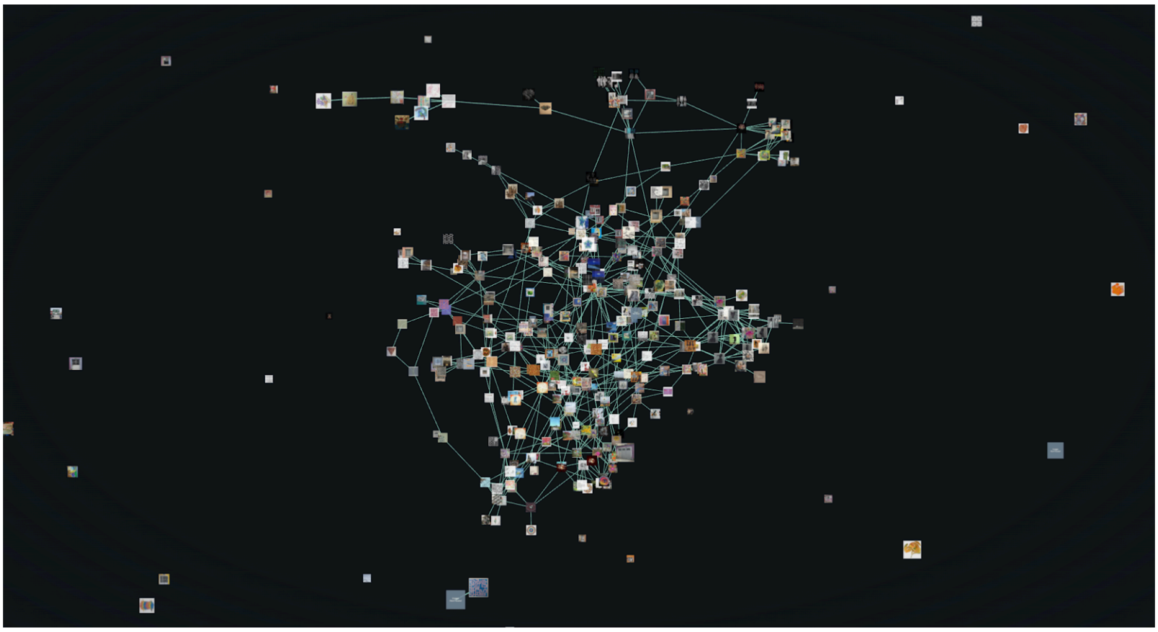
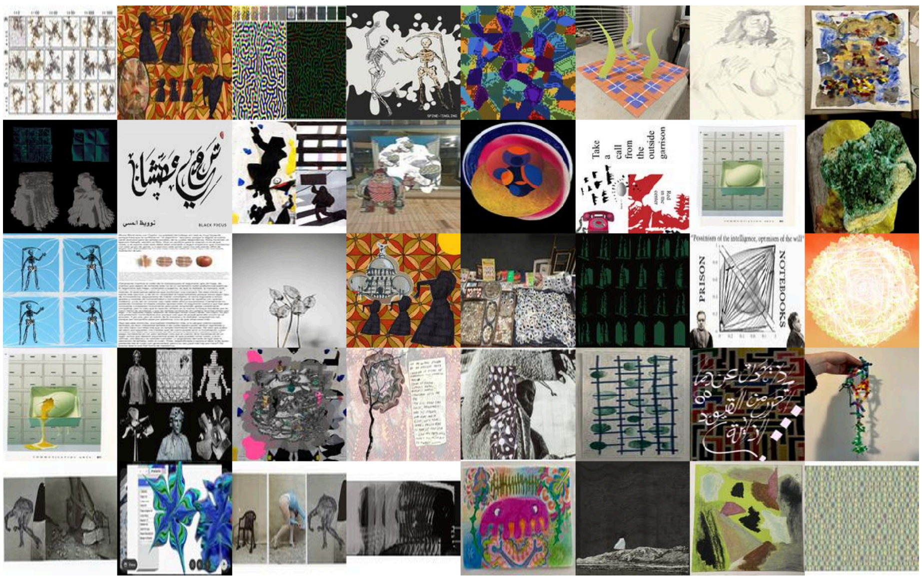
Contact: **Phileas Dazeley-Gaist**



(The sympoietic art organism citation network, from the project website)



(A random sample of artworks made for the Sympoietic Art Organism project. Motifs are repeated and reinterpreted from artwork to artwork. Can you spot the shared themes?)

Introduction

Donna Haraway, in her influential work "Staying with the Trouble: Making Kin in the Chthulucene," describes sympoiesis as a process of "making-with" rather than self-making. Unlike autopoiesis, which refers to self-producing systems that maintain their boundaries, sympoiesis emphasizes collective production, interdependence, and the ongoing collaboration among diverse entities. In Haraway's framework, sympoietic systems are open, dynamic, and continuously shaped by their interactions with other agents and environments, fostering creativity and transformation through shared agency.

This work, initiated by students at the Santa Fe Institute's 2025 Complexity Global School (CGS25) at the Universidad de Los Andes, in Bogota, Colombia, aims to explore the process of cultural production and creative collaboration in a structured but open-ended collaborative art project. Drawing on Haraway's conception of sympoiesis, we designed the project to facilitate a generative, collective creative process.

Sympoiesis, as articulated by Haraway, highlights the importance of co-creation, mutual influence, and the emergence of new forms through relational dynamics rather than isolated, disconnected individual effort. Here, we showcase the citation network grown by our small artist community in intentional collaboration working from a shared commons. Over a series of co-creative cycles, artists iteratively remixed and took inspiration from works already in the commons, contributing new pieces while explicitly tracking citation relationships. Because the network grew sympoietically, made from pieces that are themselves made from readily available materials, and organically, with artists completely free to structure and create according to their own sensibilities, we refer to the network and co-creative process as a whole as a "sympoietic art organism".

Network analysis yields insight on how this organism evolves: how artists preferentially connect to certain works, what selection and creation strategies emerge across generative cycles, and how distributed creative decisions aggregate into collective structure. The project scaffolding, making influence relationships explicit as they form, provides direct observation of the dynamics underlying cultural production and creative collaboration in a structured but open-ended participatory process.

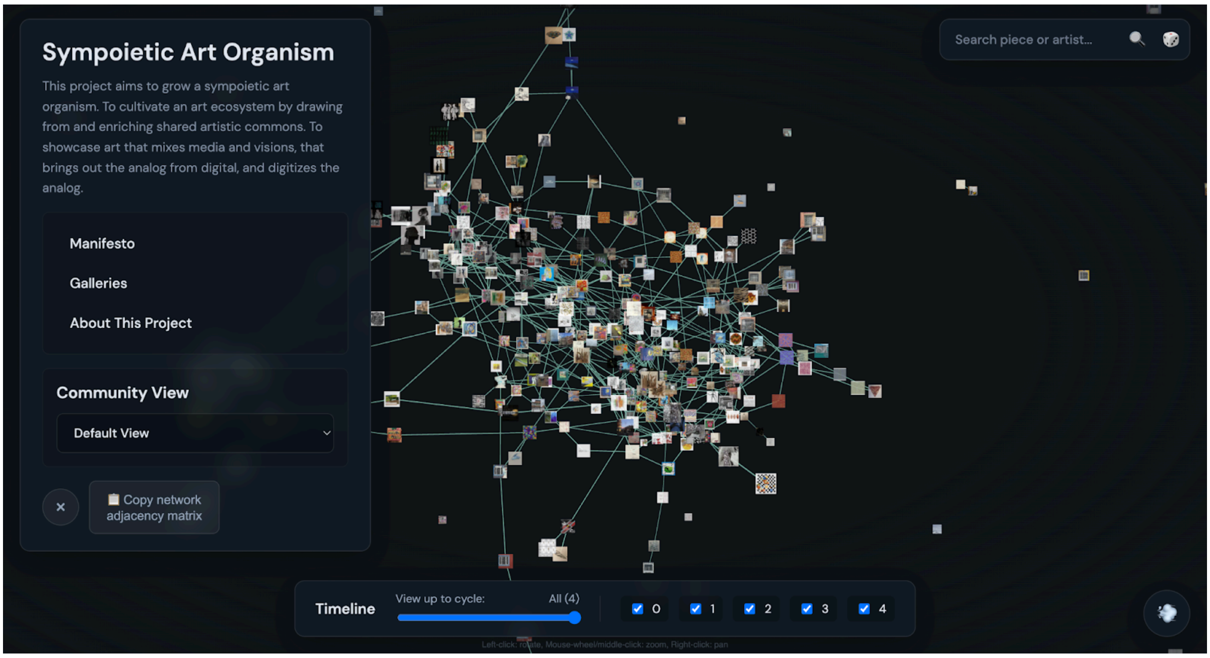
In this report, we start by reviewing the essential artefacts produced for this project. From there, we outline the methodology and planning used to structure our generative cycles, and the necessary context for the computational analysis of the final citation network that follows. The report concludes with the reflections provided by the artists themselves, and a template for readers interested in growing a sympoietic organism of their own.

Project artefacts list

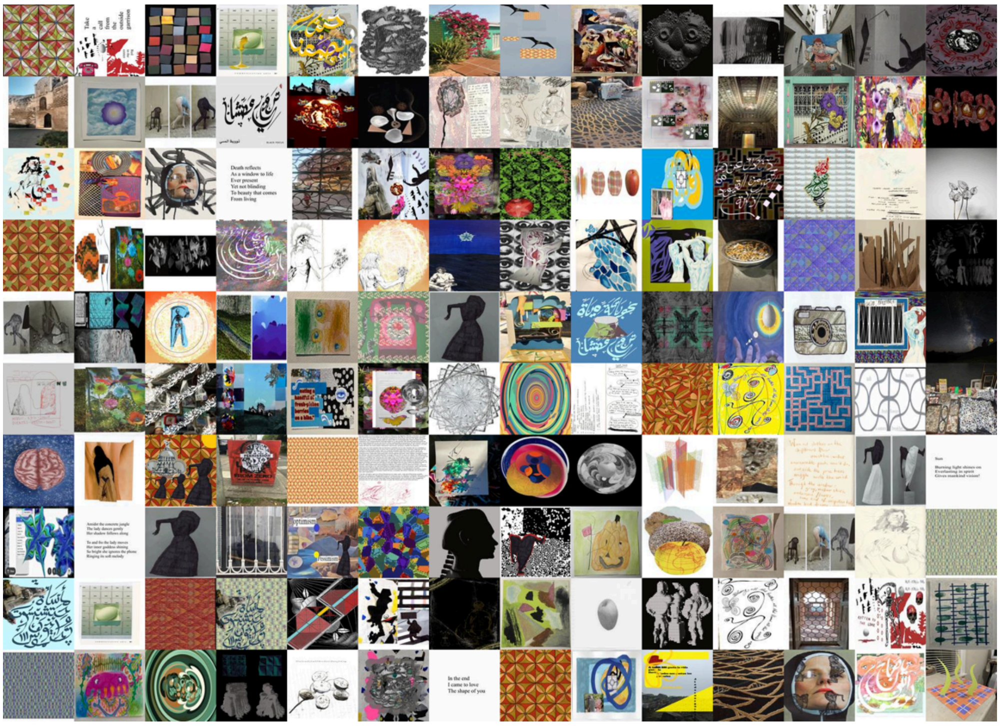
Project website:

To highlight our work, we've developed a website with which one can interactively visualize the network of citations and the artwork that make up the sympoietic art organism. The site features a network visualization tool for exploring citation relationships among artworks and digital galleries of the work of each participating artist for this project:

[Project website: Sympoietic Organism Network Visualizer]



[Project website: Digital Gallery]



Logistical and functional project documents:

Beside our website, we naturally created other documents throughout the project, some functional, to organize our process and collaborate effectively, others to share our work and ideas with a wider audience. The project planning document was written during the second week of the CGS25, and lays out the project goals and projected timeline (which we broadly stuck to). The art citation spreadsheet is among the most important and central documents in this whole project. It contains entries for each project art piece recording their immediate citation relationships with other pieces on the art network. We also link to the creative commons repository where all artworks made for this project are hosted, to a manifesto co-written by participants about the sympoietic art organism, and to the project website github repositories.

- Project planning document: A Plan to Co-creatively Grow a Sympoietic Art Organism
- Project art citation spreadsheet: Sympoietic Art Organism Shared Works Table
- Creative commons folder containing all art made for this project
- Sympoietic Art Organism Manifesto: [O.A.D.C.A.M.Q.H.L.E]
- Gallery website github repository: phileasdg/sympoietic
- Network visualizer website GitHub repository: phileasdg/sympoietic-art-organism

Presentation slides:

We have hosted presentations about this work three times since the start of this project, at the SFI Complexity Global School, at the inaugural Complexity Coffee: Nascent Research Ideas webinar (a talk series organized at the initiative of CGS alums in partnership with the Santa Fe Institute), and at the Wolfram Technology Conference.

- Talk slides: SFI CGS 2025 Project Presentation - Organic Autocatalytic Disruptive Collective Anarchist Magical Quilting: Co-creation of a sympoietic art organism
- Talk slides: October 2025 Santa Fe Institute Complexity Coffee - Nascent Research Ideas: Growing a Sympoietic Art Organism: Structured co-creativity in a small artist community.
- Talk slides: 2025 Wolfram Technology Conference: Networked Creativity: Citation Network Analysis of a Small Artist Community (there will be a recording available soon)

Methodology and planning

The project is made up of three phases, which are described below.

Phase 1: Seeding the commons

Each artist contributes a set of personal works (unfinished, polished, failed, loved, orphaned, digital, textual, visual, sonic) to a shared online drive. There is no gatekeeping. All material is welcome. All materials in the drive are treated as open to manipulation, recombination, and reinterpretation by others in the group.

Phase 2: Generative cycles

Every two weeks, each artist contributes at least five new pieces derived in whole or part from materials in the commons.

New works can:

- Integrate multiple existing pieces
- Deconstruct and fragment earlier pieces
- Translate across media (e.g., image to sound, poem to movement)

- Introduce new material, provided it builds on or relates to the commons

Every piece is labeled on the commons repository and spreadsheet using the schema:

Schema: [Cycle Number]-[Artist_Name]-[Unique Piece ID]
Example: C3-Jane_Doe-1

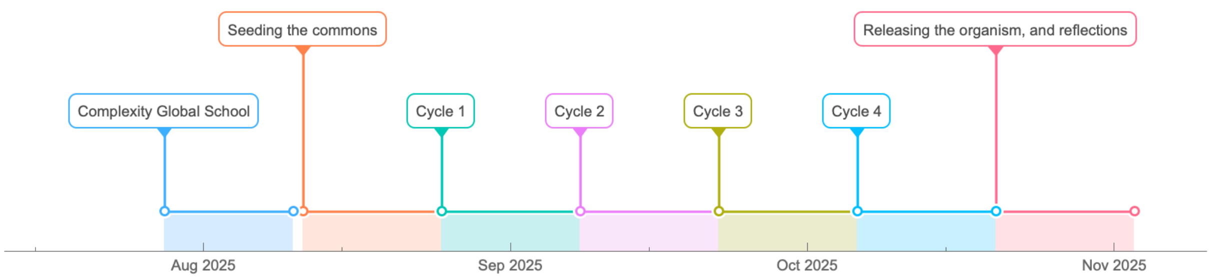
The shared metadata spreadsheet logs:

- Title/ID of each piece
- Creator
- Cycle
- Parent pieces (with IDs)
- Brief description or notes (optional)
- Type/medium (optional)

Phase 3: Releasing the organism, and reflections

The spreadsheet tracks the kinship between pieces, giving a complete genealogy: a record of how pieces evolve, combine, or give rise to others. This allows us to visualize the relational structure of the organism over time.

Project timeline



Project start: Mon 11 Aug 2025
Project end: Tue 17 Nov 2025

Seeding the commons: Mon 11 Aug 2025 to Sun 24 Aug 2025

Project Cycles:

- *Cycle 1:* Mon 25 Aug 2025 to Sun 7 Sep 2025
- *Cycle 2:* Mon 8 Sep 2025 to Sun 21 Sep 2025
- *Cycle 3:* Mon 22 Sep 2025 to Sun 5 Oct 2025
- *Cycle 4:* Mon 6 Oct 2025 to Sun 19 Oct 2025

Licensing and ethics

All shared materials are considered to be under a Creative Commons Attribution-ShareAlike (CC BY-SA) license by default.

Artists must credit derivative works using the IDs of source pieces.
No one “owns” the final form; all contributions remain traceable but are shared.

Computational analysis of the art citation network

From the project art citation spreadsheet, we can parse the reported citations for each piece to construct a network representing all known relationships between artworks. In this network, each art piece is a node, and an explicit citation relationship is a directed edge pointing from the child (the new derivative work) to the parent (the source piece). This structure describes the complete explicit genealogy of artworks from the sympoietic art organism, allowing us to characterize and study the structural patterns of creative influence that emerged during the generative cycles.

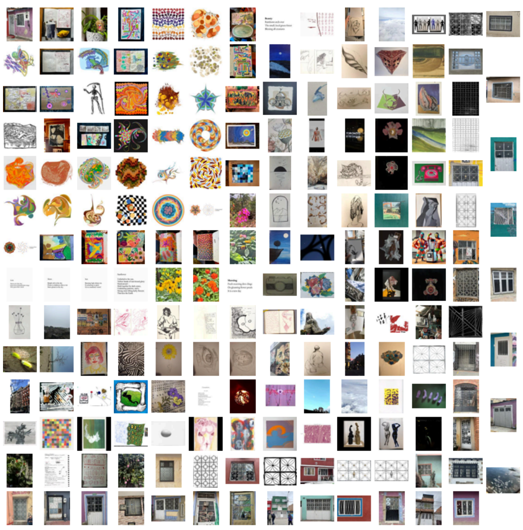
We explored the patterns underlying this sympoietic process using a variety of computational methods, including: network visualizations, analysis of degree distributions and community structure, node centrality measures, co-citation histograms, and artist-centric citation subgraphs.

Developmental analysis of the art organism citation network

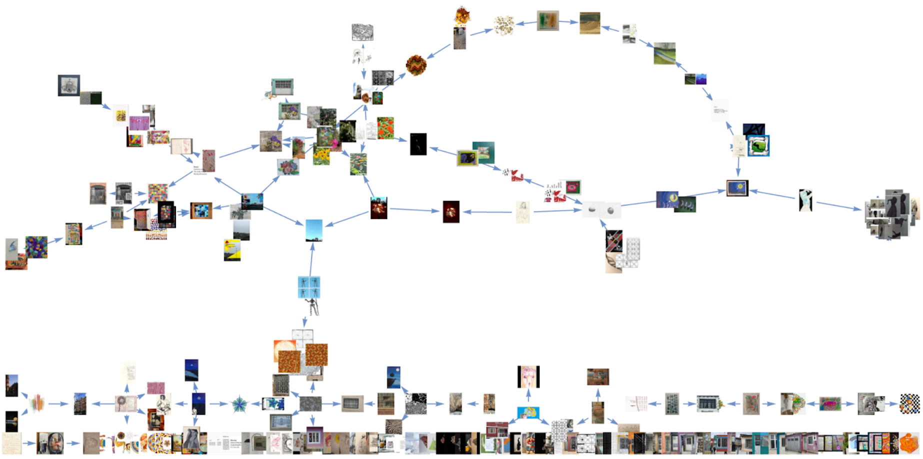
Network structure

The figures that follow give a visual account of the organism's growth.

Seeding phase: During the seeding phase, artists contributed 10 pieces each. These are not linked in the citation network. Here they are represented as an image grid:

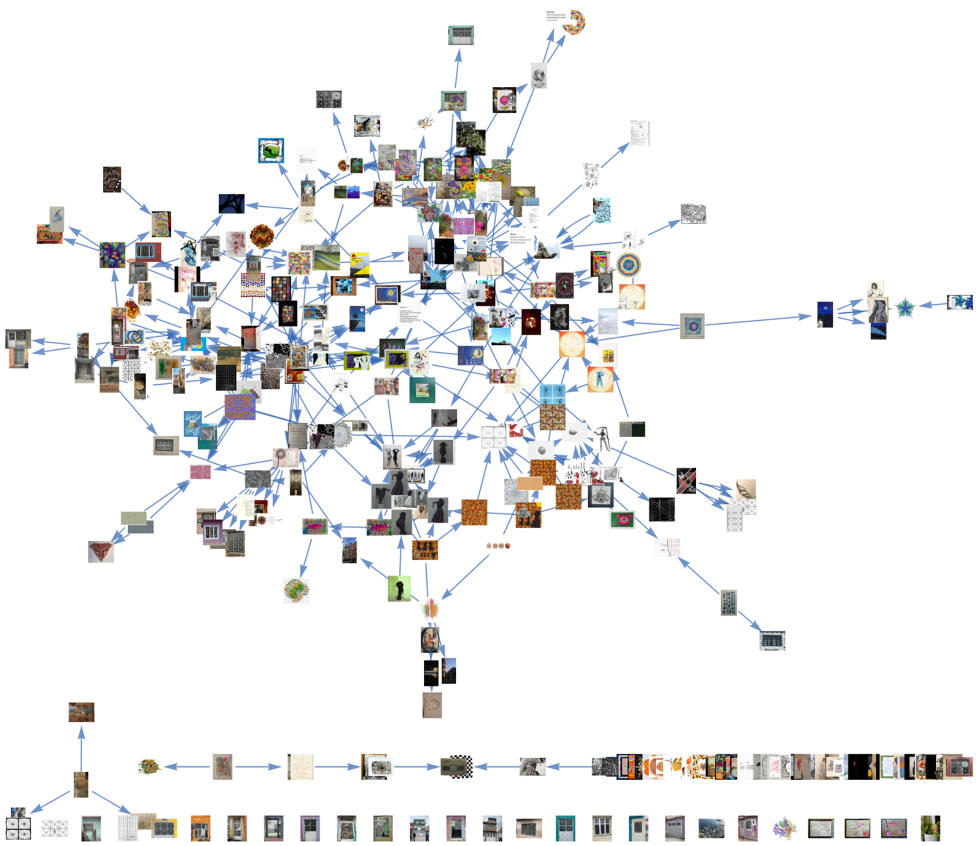


Cycle 1 graph structure: At the end of the first cycle, the art citation network was like this:



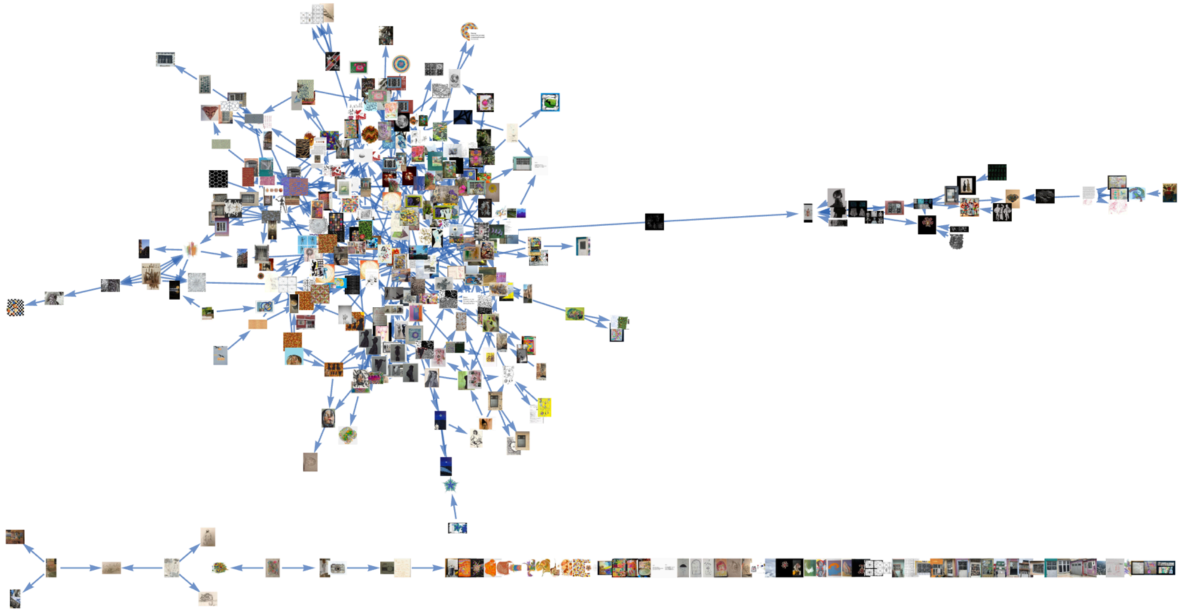
In the main connected component of the graph (MCC), co-citations formed chains, some of which made undirected cycles. This happens because several cycle 1 pieces share the same sources. When two pieces share a single source, they make up a link in the chain. When a piece shares a different single source with two other pieces, they make up two connected links in the chain. The chain path is closed if any two pieces in the sequence are connected back through their shared citation of a common seeding phase node. Since cycle 1 nodes cannot cite each other, this shared ancestry is the only mechanism creating closed paths when the directionality of the citation is ignored.

Cycle 2: By the end of the second generative cycle, the graph looked like this:



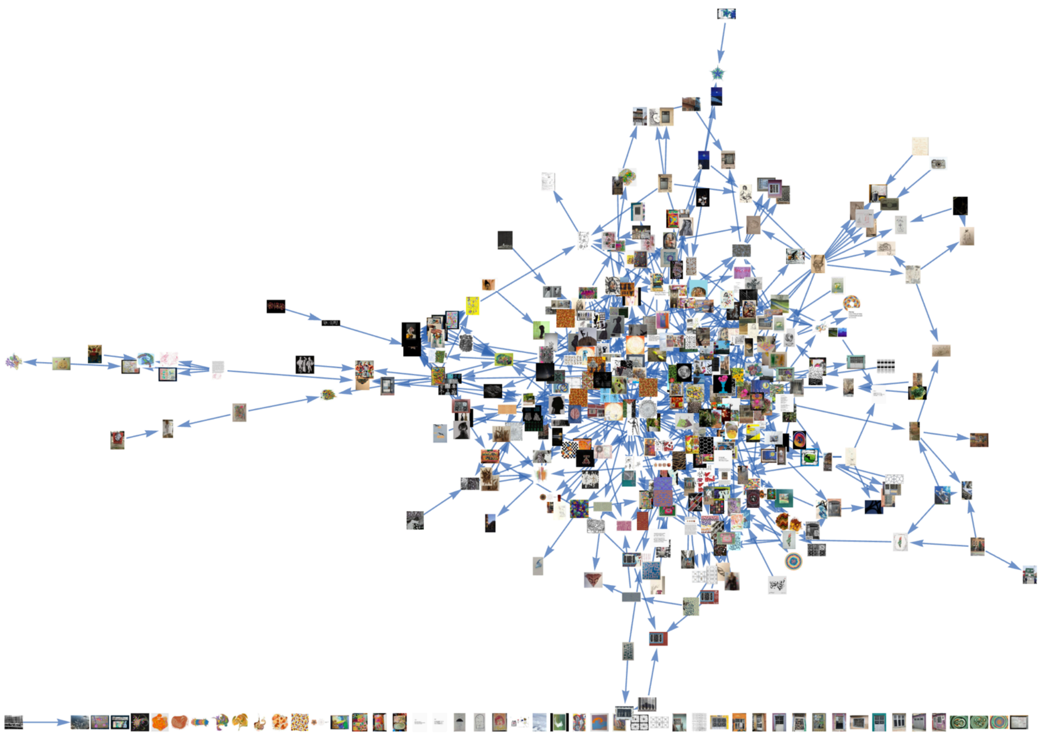
The main connected component is larger, more tangled, and has swallowed some of the smaller connected components from the cycle 1 graph, though there are still small connected components and disconnected nodes floating about.

Cycle 3: By the end of cycle 3, the network had grown a tail:



The graph's MCC has a long tail extending to the right. It is made of overlapping co-citation chains, and connects to the body of the MCC in a single point. This tells us that despite being connected to the MCC, the tail pieces and the pieces in the MCC core are largely structurally isolated from one another.

Cycle 4: By the end of cycle 4, the network had reached its final form:



The graph is strongly centralized. The tail that was visible at the end of cycle 3 has been absorbed into the MCC, which dominates the visualization. The largest connected component besides it is made up of only two nodes.

The table below summarises some of the global structural properties of the citation network at each stage of the project. From the Vertices and Edges columns, we can see that the network experienced continuous growth and a sharp increase in structural complexity beginning in cycle 2, reflected in the shift from zero to non-zero clustering coefficients (CC).

Project Stage	Vertices	Edges	Density	Global CC	Mean CC
Seeding phase	190	0	0	0	0
Cycle 1	251	178	0.002821	0	0
Cycle 2	307	375	0.003971	0.04038	0.02647
Cycle 3	362	521	0.003971	0.0319	0.02071
Cycle 4	411	725	0.004291	0.03664	0.03973

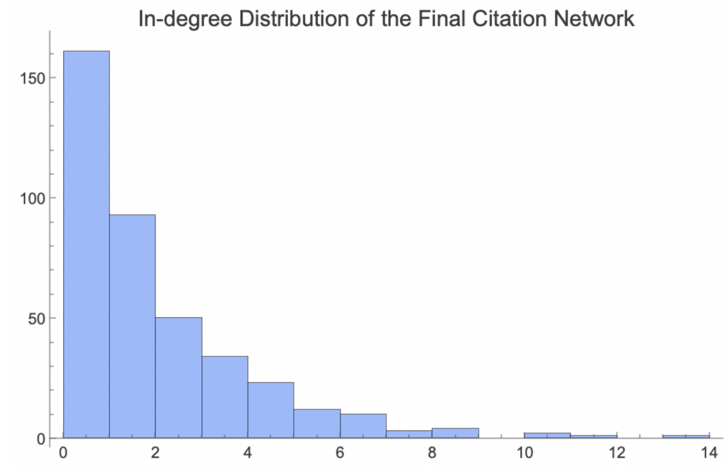
We report clustering coefficients for the undirected graph; these coefficients are zero in the directed network due to its predominantly directed-acyclic structure.

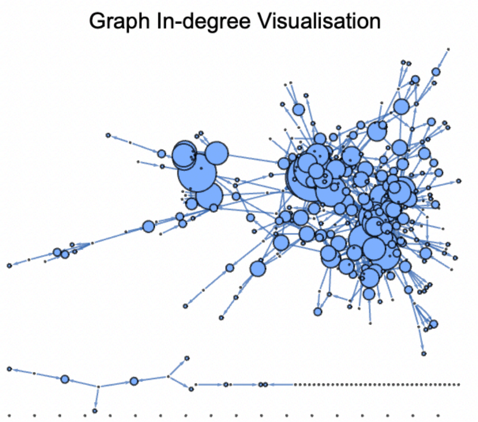
From cycle to cycle, the graph showed fast structural consolidation into a dominant central core by cycle 4. This consolidation is reflected in rising clustering coefficients, which show the emergence of convergent citation patterns (artists were more and more influenced by overlapping sources) and local structural redundancy in the core (the core of the final network is highly interwoven because many pieces share common ancestry).

Centrality and influence

In-degree distribution

To quantify influence, we analyzed the network's in-degree distribution (citations received, *child → parent*):

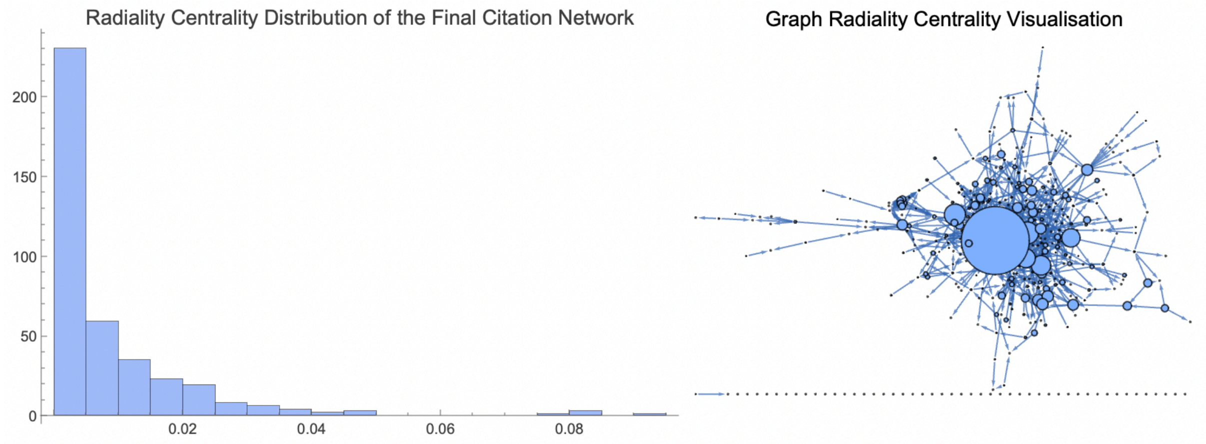




The distribution shows a pronounced positive skew with a rapidly decaying slope and a long, heavy tail. This shape strongly suggests the network has scale-free characteristics, meaning influence is highly centralized and governed by a "rich-get-richer" dynamic where a few emergent hub pieces dominate citation flow. Approximately 35% of works received zero citations, acting as creative dead ends.

Radiality distribution

We computed the radiality centrality distribution of the final graph to quantify how structurally integrated artworks are within the network. Radiality measures a node's efficiency in communication paths relative to all other nodes, reflecting the inverse of its average distance to all other pieces in the network..

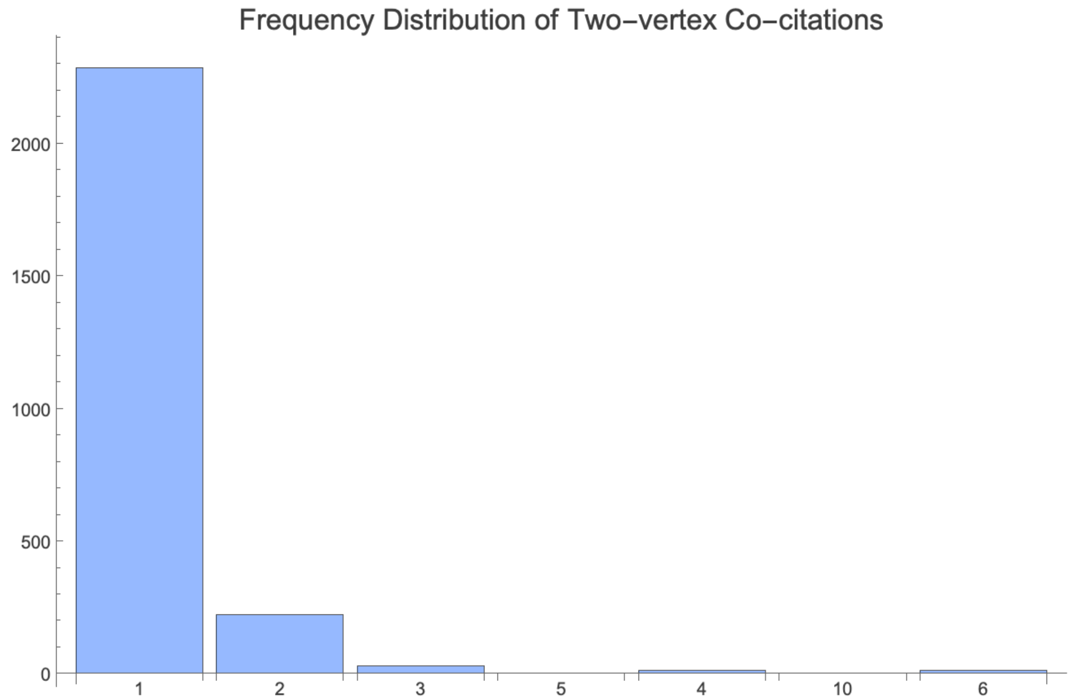


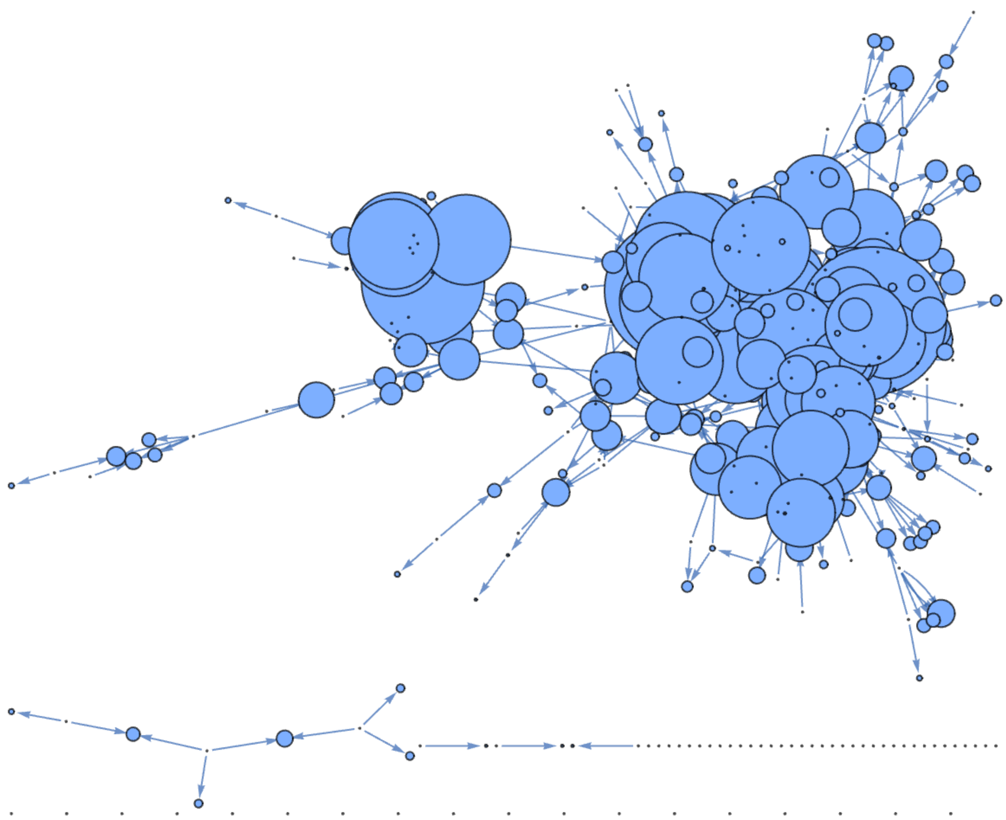
The distribution is highly concentrated near the origin, with over 200 works having very low radiality (radiality < 0.005). The vast majority of works are structurally peripheral to the main flows of influence. However, the accompanying graph plot visualization clearly shows hubs (large blue nodes) with high radiality. The top three nodes with the highest radiality are 3-Nadine_AbdElRazek-1 (0.09), 4-Arango_Franco_Sara-4 (0.082), and 2-Phileas_Dazeley_Gaist-1 (0.081).

Two-piece co-citation distribution

We also analyze the frequency distribution of two-vertex co-citations to quantify the overall prevalence of creative partnerships within the network. Specifically, this plot measures how often any given pair of source artworks was cited together in a new derivative work. The distribution is extremely skewed, with most co-citation events (approximately 2,300 pairs) occurring only once. This tells us that artists strongly favored novelty in their pairings, rarely repeating the same combination of sources. However, the accompanying graph visualization, where node size is scaled by the total number of times a piece was part of *any* co-citation event, shows that most nodes within the MCC are large.

Read together, the plots show that while specific pairings were rarely repeated, the act of co-citation was common, and the same central nodes were consistently re-used in new, novel combinations.

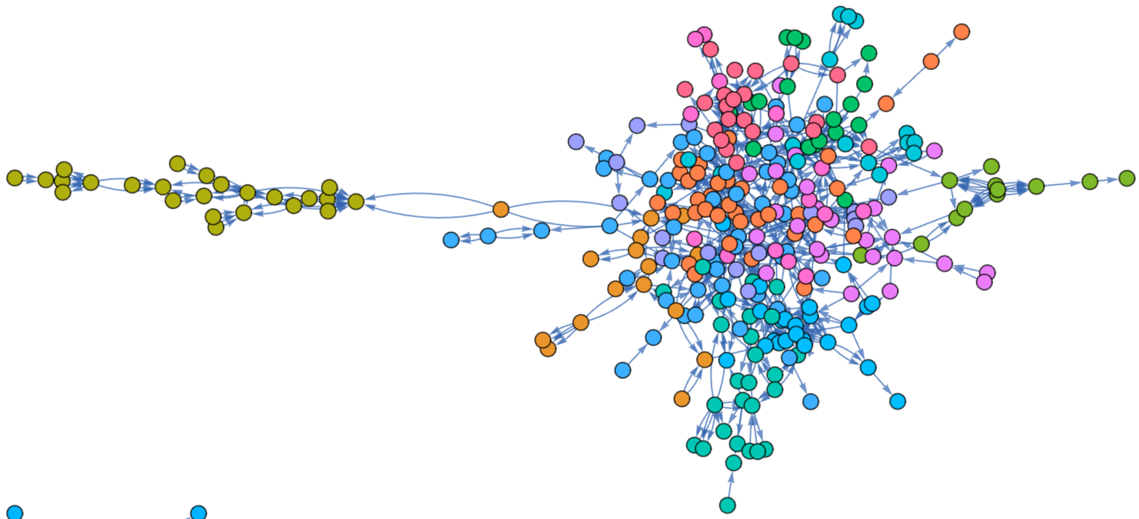




Creative clustering

Community detection

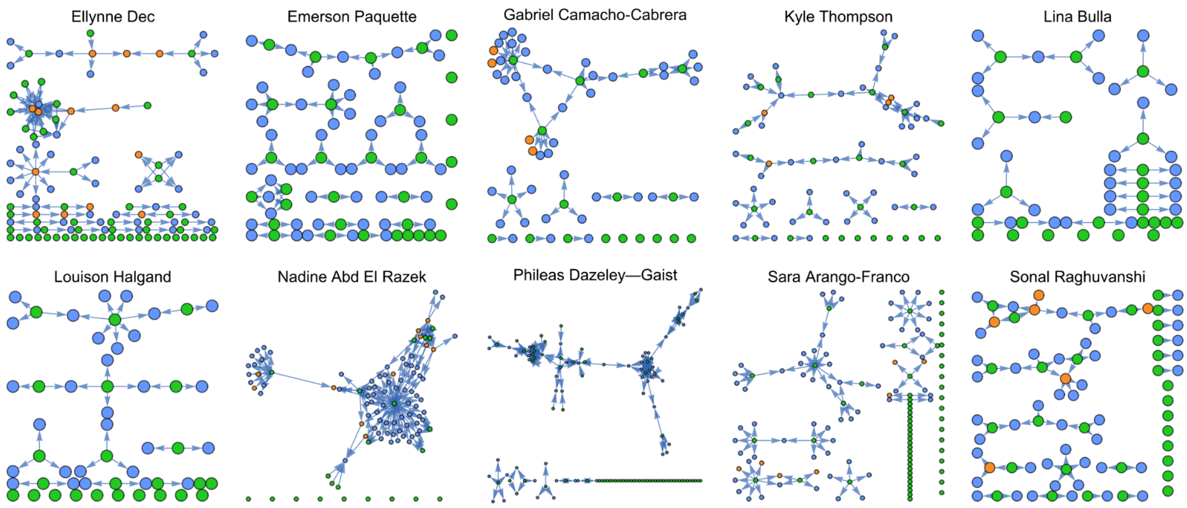
To identify emergent structural sub-groups, we applied community detection using modularity maximization. The plot below visualizes the final network, with nodes colored according to their assigned community.



The community detection analysis reveals two distinct structural patterns. The MCC is a highly integrated mix of all major communities (pink, teal, orange, purple), suggesting that the core might be a collaboration between structural groups, not a monolith. In contrast, the peripheral components (like the large "tail" on the left, colored olive green) are monochromatic, suggesting that they are structurally isolated, internally-focused sub-communities.

Artist-centric citation subgraphs

Next, we analyze artist-centric subgraphs: subgraphs consisting of an artist's works and the pieces they cite to see if these clusters are driven by individual creative habits. In the plot below, green nodes are artist submissions, blue nodes are works that were cited by the artist in focus, and orange nodes are self-citations.



The subgraphs in the plot above reflect differences in the creative strategies of the contributing artists. The network structure highlights differences in integration style reflective of artist choice. Take, for instance, assortativity. Nadine Abd El Razek's strong negative assortativity (–0.312) is indicative of her role as a large-scale integrator, linking many simple units to central hubs.

Conversely, Sara Arango-Franco's positive assortativity (0.215) shows a "hub-to-hub" strategy, reinforcing existing strong ties. Furthermore, high component counts (e.g., Ellynne Dec, 109) results from artists making many simple, separated lineages by citing non-overlapping pieces between their own art submissions.

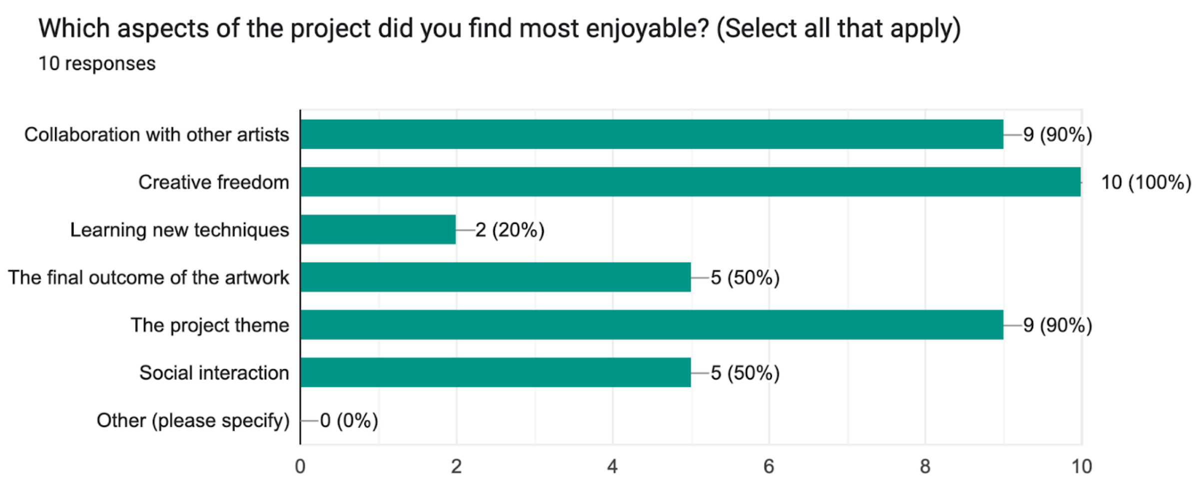
	Connected component count	Acyclic?	Undirected acyclic?	Graph density	Graph degree assortativity (homophily)
Ellynne Dec	109	True	False	0.00781515	−0.154627
Emerson Paquette	81	True	False	0.00910494	0
Gabriel Camacho−Cabrera	62	True	False	0.0132205	0
Kyle Thompson	72	True	False	0.0113459	−0.165919
Lina Bulla	53	True	True	0.0112482	0
Louison Halgand	53	True	True	0.0119739	0
Nadine Abd El Razek	104	True	False	0.012696	−0.312401
Phileas Dazeley—Gaist	127	True	False	0.00781152	0.214937
Sara Arango−Franco	127	True	False	0.00487439	0
Sonal Raghuvanshi	82	True	False	0.00918398	0.138982

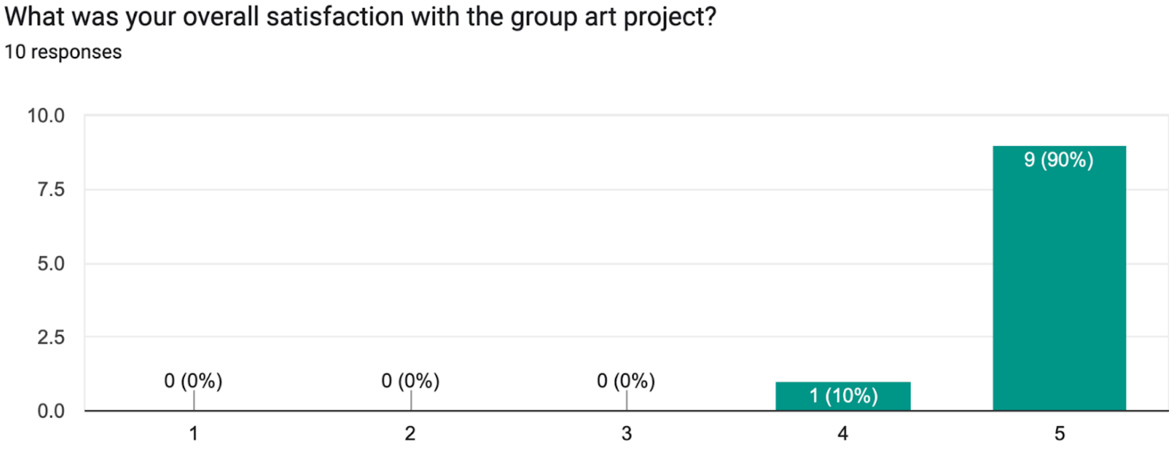
Reflections and perspectives from the participants

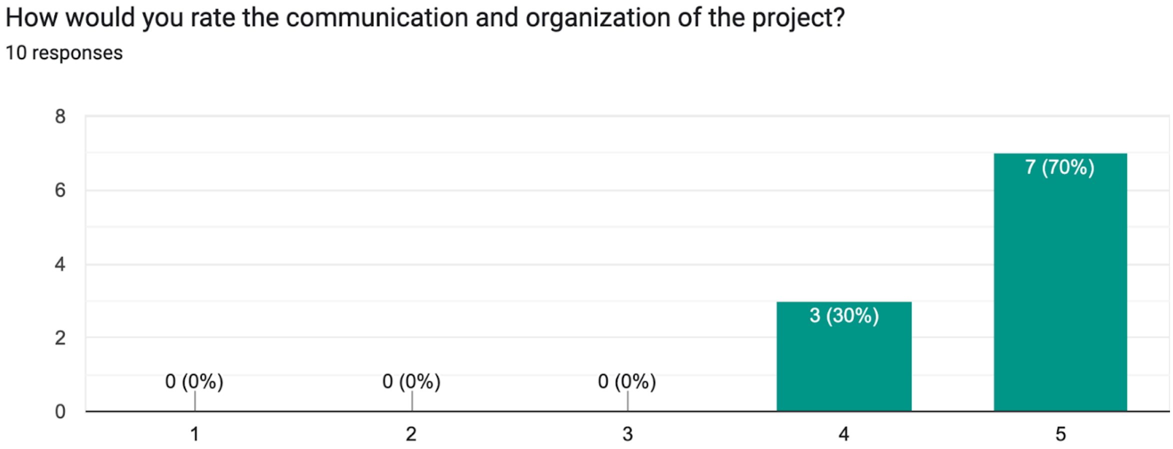
The participant feedback, collected via questionnaire, paints a detailed picture of the … Sympoietic Art Organism project. This section will present a qualitative and quantitative summary of that feedback, with bar charts summarizing the responses to the feedback questionnaire's checkbox and scoring questions alongside the presented written reflections. The reflections presented in this section are supplemented and substantiated by the computational emotion analysis of participant feedback detailed in *Appendix II*.

Overall experience

The participants consistently reported a positive and meaningful experience, describing it as "exceptionally enjoyable," "very fun and inspiring," and a "comforting, engaging, beautiful" effort. For many of us, it provided a kind of self-discipline. It was a "great exercise to just make the time" for creation, which helped us practice "not judging [ourselves] too harshly" and focus on the action of making art rather than overthinking the result.





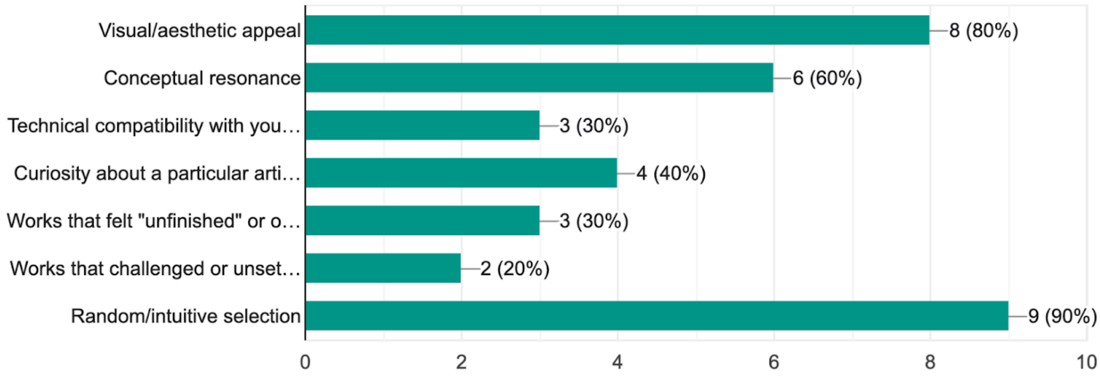


Working with the commons

As the commons grew, the participants used different strategies to choose and connect with each other's work. Most reported picking pieces at random, intuitively, or based on their visual/aesthetic appeal. Artists reported allowing pieces to "find them," guided by an "overall vibe," color, or specific natural elements. One participant reported feeling a creative tension between wanting to maximize the diversity of authors referenced and the natural tendency to be "drawn back to certain pieces,". Another described their search process as being reflective of "where my aesthetic taste is today." Notably, the increasing visual complexity in later cycles prompted some participants to deliberately look back to engage with materials from the project's seeding phase and early cycles.

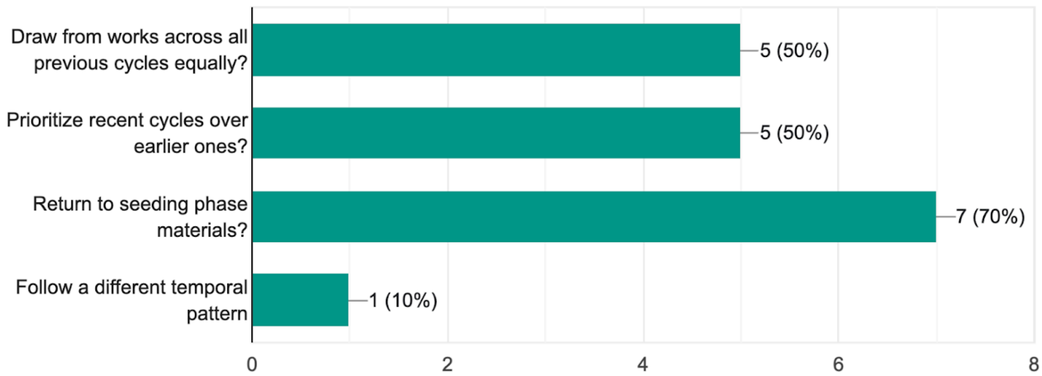
When choosing which works to engage with, what factors mattered to you? (Check all that apply, and feel free to elaborate)

10 responses



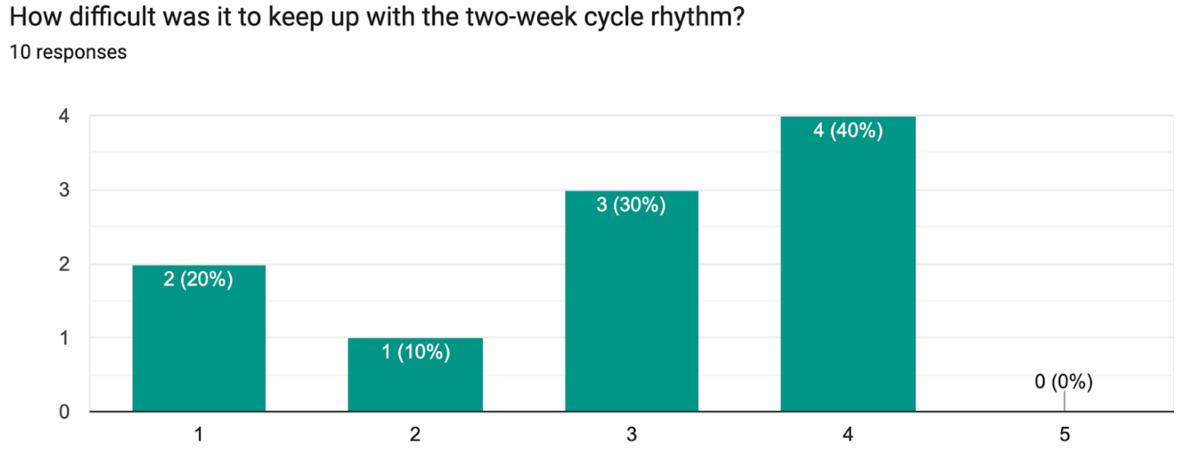
In your practice, did you:

10 responses



Time, process, and creative approaches

The two-week cycle rhythm was the project’s greatest logistical challenge, with time management being the primary difficulty for the participants. The nature of the project forced participants to find ways to manage the tension between our abilities to contribute polished “finished” works, and our time



Relationality and emergence

The heart of the project was the collective dialogue among the artists. Participants experienced working with others' contributions as a "really nice and fruitful" opportunity to engage in a visual dialogue with people they did not always personally know. One artist remarked that it was a "very specific way of getting to know people."

When asked to describe the qualities, moods, and character of the project, participants described it using words like “tangled”, “shy”, “musical”, “poetic”, “dancing”, as well as “self-sustaining” and “entropy-increasing”.

Challenges and improvements

Participants sometimes faced operational challenges related to time management, the "intensity" of switching rapidly into creative mode, and the logistical friction of art piece organization, navigation, and contribution in the growing commons. The operational recommendations from participants for future projects like our own included extending the cycle rhythm to be monthly and implementing a dedicated file submission system (or an alternative to Google Drive) to improve material navigation and data consistency.

Looking forward

We are highly enthusiastic about the future of this project and others like it. Indeed, we hope others will consider trying to grow their own sympoietic organisms. We have created a template for others seeking to do so in which we list the elements and design considerations we feel are most important for success (Appendix I). We are also interested in potential pedagogical applications of this project design.

Appendix I: Template for future sympoietic art organisms

In an effort to replicate the organism of artistic co-creation, we invite other creatives to use the seeds here as a template to grow their own artistic art organism. Our intended audience is intended for any group of people dedicated to creating intentionally with others in their own community or around the world. This template was created to be a participatory modeling framework that can encourages creativity and spontaneity.

Title

Growing a Sympoietic Art Organism

Project Background

The goal of this project is to grow a **sympoietic art organism**. Two things are of note in the creation and growth of a sympoietic art organism:

1. Cultivating an art ecosystem by drawing from and enriching shared artistic commons.
2. Showcasing art that mixes media and visions; art that brings out the analog from digital, and digitizes the analog.

By **sympoietic art organism,** we mean art that can be thought of as an artwork (or more precisely, an ongoing artistic process) that is poetic, conceptual, and ecological in nature. It is:

- Co-created (not made by a single author),
- Open-ended and adaptive (always becoming),
- Responsive to its environment (not closed off from change),
- Distributed in agency (not centered on human control), and
- Relational (interdependent with other systems, beings, or forces).

The word “sympoietic” comes from sympoiesis, coined by systems theorists and developed by Donna Haraway, meaning “making-with” rather than autopoiesis, which is self-making. Sympoiesis is collective, dynamic, and contingent.

Project Objectives (Explicit and Implicit)

Explicit: To create an emergent network structure of artworks that evolve over time, building off each other in a deliberate, collectively shared creation process.

Implicit: To understand how stem education can be incorporated into artistic expression and creation

Core Modeling Personnel

For this project, there should be an individual or group of individuals dedicated to tracking, maintaining, and constantly transforming the art organism during the growth process. It’s also possible for tasks to be split up as needed based on mutual agreement between artists. The role of the modeling personnel includes:

- A person (or several people) with skills in network visualization software such as Python, RStudio, Wolfram Language.
- A mechanism for tracking all explicit artistic contributions and relationships between them, and a repository in which to store artworks in the commons.

    - We recommend coming up with a naming scheme that uniquely identifies each art piece. For example the snake-case scheme: [Cycle Number]-[Artist_Name]-[Unique Piece ID]
- The modeling team or modeler will maintain ongoing communication with participants and coordinate the project.

Releasing the Organism

We suggest finding a platform on which to release the art organism to the world. Depending on the skills of your group, you could publish results in GitHub, on a website, as a written essay with different sections split up between artists, as a book or binder of works to be shared around, etc.

Participants

Anyone can participate in this project. Individuals must be willing to co-create with other artists in a meaningful, constructive way for an agreed upon amount of time. **We recommend splitting up project cycles weekly to maintain dedication and consistency.**

Project Cycles:

- *Cycle 1:* (insert dates here)
- *Cycle 2:* (insert dates here)
- *Cycle 3:* (insert dates here)
- *Cycle 4:* (insert dates here)

Process

The instructions for this section borrow from Fluxus artists, who created instructions for performance and visual art. Think of each part of this process as a piece of art on its own; something that is curated and developed by individuals working in collaboration with each other.

Exquisite Corpse Exercise: This exercise is meant to provide a structure for the naming convention of your particular art piece.

Every artist should sit in a circle facing each other.

Going clockwise, say a word that conjures your aesthetic mode in the moment.

When you come round to the first person, you have a corpse. Isn't it lovely?

Take your word and write a paragraph.

What do you feel?

Come together with your artist collective

Piece every individual word and paragraph alongside each other

Now, the corpse is alive!

Prompts for Creating Pieces (Feel free to add any that you have tried as well)

Prompt 1:

Think about an experience during your day

Capture it in your mind and soul

Release it as a poem, sound, or collage

Prompt 2:

Look at a fellow participant's art piece. Write a short sentence (5-7 words) on what the piece makes you feel, think or want. Look at another person's art piece that is similar in mood to the first sentence. Add another statement onto that. Repeat until you are satisfied.

Tips on how to manage data gathered

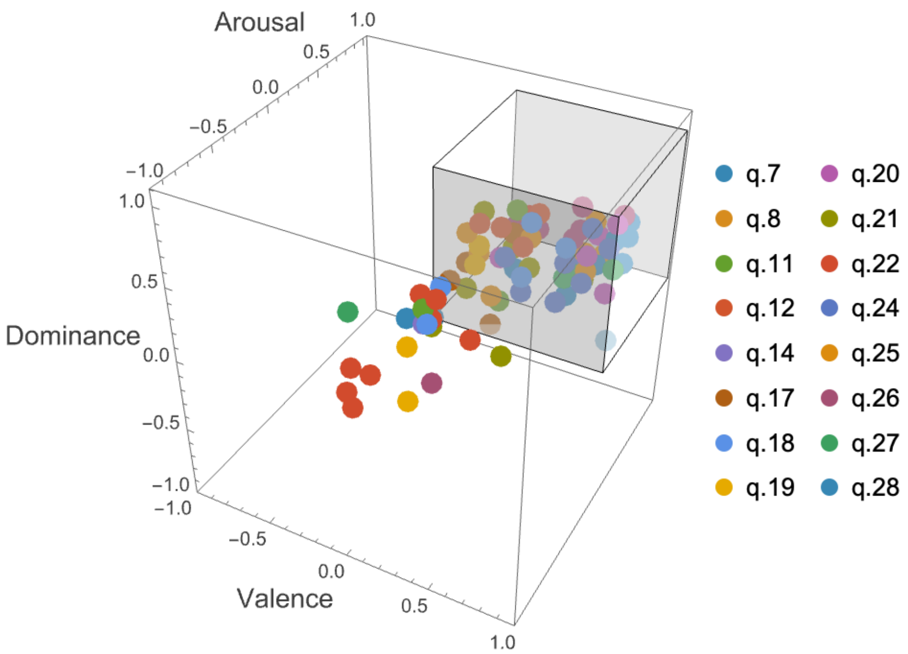
Ensure that all pieces have the same formatting throughout the creation process, so nothing gets mixed up. A piece can have different names (e.g. 0-Smith_Joe-1, or 0-Smith_Joe-Flowers), but the numbering and naming convention has to be consistent. Use a central, shared space like google drive to hold all the works together. Individuals who are conducting the network modeling should check before the end of an art cycle to ensure the participants have uploaded their work properly.

Appendix II: Participant feedback emotion analysis using natural language processing

To supplement the qualitative analysis of participant feedback of this last section, we also performed a computational emotion analysis using Anthropic's Claude Sonnet 4.5 to place de-identified participant feedback questionnaire answers on Valence Arousal, and Dominance (VAD) scales defining each in accordance with the spatial model of emotional state introduced by Albert Mehrabian and James A. Russell[1]. To the extent that the language model succeeds at classifying the emotional contents of the participants' answers, and to the extent that human emotions can be mapped onto the dimensions of the VAD model, this process allows us to model the emotional contents of the feedback overall.

The questions we considered in the following analysis can be found in *Appendix III: Feedback questionnaire*, where they are numbered 7, 8, 11, 12, 14, 17, 18, 19, 20, 21, 22, 24, 25, 26, 27, and 28, and highlighted in green. These correspond to the list of questions that required written responses, excluding those that pertain to website gallery set-up details. They were supplied programmatically to the classifier model, alongside a prepended prompt which can be found in *Appendix IV: Participant feedback emotion analysis language model prompt*. The results are presented below.

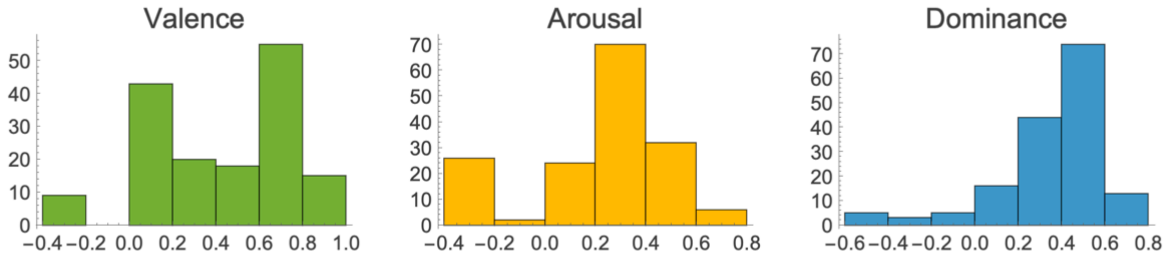
Participant feedback represented in affective (VAD) space: Each point corresponds to an answer scored automatically for valence, arousal, and dominance. Approximately 72% of results scored positively for all three dimensions of the model, in the first octant (represented in the plot by the unit cube). This region of affective space is typically associated with strong, positive, and activating emotions where the person feels in control. Examples of emotions that fall firmly in this region include joy, happiness, excitement, admiration, enthusiasm, satisfaction, and triumph (Bradley & Lang 1999)**.**



[1] Also called the Pleasure Arousal Dominance, or PAD emotional state model (Mehrabian, 1980).

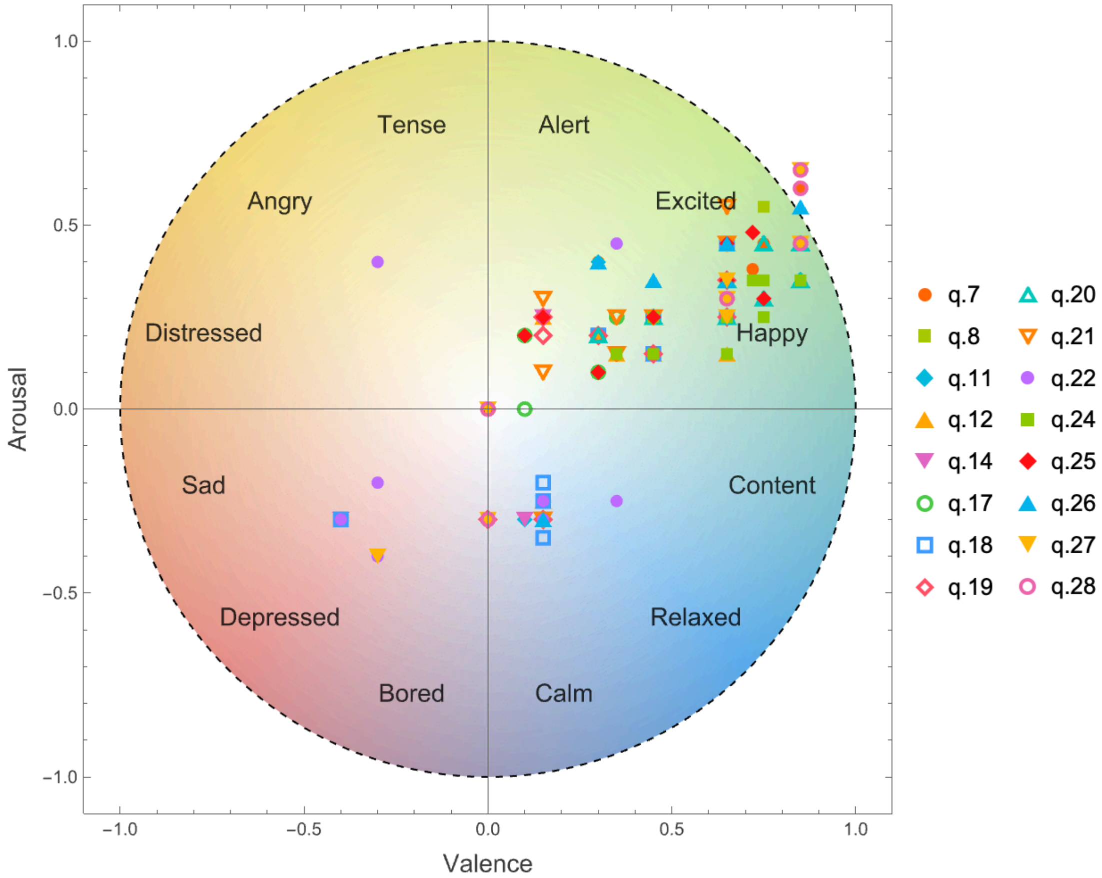
Valence, arousal, and dominance score distributions in participant feedback:

The distributions of participant feedback VAD scores each have a strong positive skew. Approximately 90% of the responses scored positively for valence and dominance, and approximately 80% scored positively for arousal. **Valence:** The distribution of valence scores is bimodal, with the highest peak in the range 0.6 and 0.8, and second highest between 0.0 and 0.2 corresponding to moderately positive affective states, and neutral to very slightly positive states, respectively. **Arousal:** The distribution of arousal scores is also bimodal, with the primary mode between 0.2 to 0.4, and secondary mode between -0.4 and -0.2, which indicate moderately intense, excited states, and moderately calm states, respectively. **Dominance:** The distribution of dominance scores is highly concentrated, peaking between 0.4 and 0.6, indicating consistent feelings of moderate to high control.



These results show that the vast majority of responses indicate **positive or pleasant** affective states characterised by a strong sense of control, and which are typically associated with a feeling of moderate intensity, or slightly less frequently, a sense of calm or relaxation.

Mapping responses onto the circumplex model of emotion experience: The first two dimensions of the VAD model, valence and arousal, were developed into another very popular model of emotional classification: James Russel's circumplex model of emotion experience (Russel, 1980), which organizes emotions into a circular space. Projected into this space, the participant VAD scores mostly cluster in the upper-right quadrant, (positive valence and arousal), corresponding to emotions like "excited" and "happy". A large majority of data points reflect pleasant, activated states, with very few observations falling into the unpleasant (left) side of the model (e.g. distressed, sad, angry). The data form two clusters in the first quadrant, differentiating between highly activated points and a small cluster of low-arousal, pleasant points confirming the bimodal distribution observed in each dimension.



In summary, the computational emotion analysis of participant feedback reveals a consistently positive emotional landscape, with most responses expressing states of high pleasantness, strong control, and moderate activation. These findings strongly suggest that the written feedback is overwhelmingly associated with positive and affirming emotional experiences related to Sympoietic Art Organism project.

Appendix III: Feedback questionnaire

1. Email address
2. Please enter your full name as you would like it to be displayed on your gallery page on the website
3. If you'd like to include any text on your website gallery page, please enter it here (for example, a blurb about yourself)
4. If you would like to highlight any specific pieces of your own on your gallery page, please list them here by their IDs.
5. If you would like any of your pieces not to appear on the gallery page, please list them here by their IDs.
6. If you are happy to have all your pieces appear, please specify whether they should appear in chronological, reverse-chronological, or random order.
    - Chronological
    - Reverse-chronological
    - Random
    - Other
7. How would you describe your experience participating in this project?
8. What moments, discoveries, or surprises stand out to you from the process?
9. What was your overall satisfaction with the group art project? (Rating: 1-5)
10. Which aspects of the project did you find most enjoyable? (Select all that apply)
    - Collaboration with other artists
    - Creative freedom
    - Learning new techniques
    - The final outcome of the artwork
    - The project theme
    - Social interaction
    - Other (please specify)
11. As the shared pool of materials expanded over the cycles, how did you navigate it? Did the growing volume change how you browsed, selected, or engaged with pieces?
12. How did you typically decide which pieces from the commons to work with?For example: Did you browse randomly? Follow threads from specific artists? Respond to particular media or themes? Let pieces find you?
13. When choosing which works to engage with, what factors mattered to you? (Check all that apply, and feel free to elaborate)
    - Visual/aesthetic appeal
    - Conceptual resonance
    - Technical compatibility with your practice
    - Curiosity about a particular artist's work
    - Works that felt "unfinished" or open
    - Works that challenged or unsettled you
    - Random/intuitive selection
    - Other (please specify)
14. Did you intentionally try to avoid using the same sources repeatedly, or did you find yourself drawn back to certain pieces? How did that pattern feel to you?
15. In your practice, did you:
    - Draw from works across all previous cycles equally?
    - Prioritize recent cycles over earlier ones?

- Return to seeding phase materials?
- Follow a different temporal pattern
- Other (please specify)

16. How difficult was it to keep up with the two-week cycle rhythm? (Rating: 1-5)
17. What methods, techniques, or processes did you explore for creating your pieces?
18. Did your relationship to "finished" work shift during the project? How did you navigate the tension between polish and participation?
19. As you looked at the genealogy/metadata, what patterns or relationships surprised you? Did you notice unexpected lineages or convergences?
20. How did you experience working with other artists' contributions? Did particular collaborations (direct or indirect) feel especially meaningful?
21. If you were to describe the art organism that emerged, what qualities, moods, or characteristics would you name?
22. What challenges did you encounter? (These could be logistical, creative, emotional, conceptual, or otherwise)
23. How would you rate the communication and organization of the project? (Rating: 1-5)
24. What aspects of the structure, process, or ethos should be preserved in future iterations?
25. What would you change, add, or remove to better support sympoietic art-making?
26. Are there pieces, relationships, or questions from this project that you'd like to continue exploring?
27. Is there anything else you'd like to share about your experience, the organism, or the process of collective art-making?
28. If you have any additional comments or feedback, please share them here.

Appendix IV: Participant feedback emotion analysis language model prompt

For the computational emotion analysis of participant feedback, each participant answer was supplied to the classifier language model with the following prompt text prepended:

> *"Analyze the overall emotional tone of the following text and assign it coordinates in VAD space, where:*
>
> *- Valence = positive to negative emotion (maximum displeasure = -1, maximum pleasure = 1)*
> *- Arousal = intensity/activation level (complete nonarousal = -1, maximum arousal = 1)*
> *- Dominance = the controlling and dominant nature of the emotion (maximally submissive = -1, maximally dominant = 1)*
>
> *The Pleasure-Displeasure Scale measures how pleasant an emotion may be. For instance, both anger and fear are unpleasant emotions, and score high on the displeasure scale. However, joy is a pleasant emotion.*
> *The Arousal-Nonarousal Scale measures how energized or soporific one feels. It is not the intensity of the emotion – for grief and depression can be low arousal intense feelings. While both anger and rage are unpleasant emotions, rage has a higher intensity or a higher arousal state. However boredom, which is also an unpleasant state, has a low arousal value.*
> *The Dominance-Submissiveness Scale represents the controlling and dominant nature of the emotion. For instance, while both fear and anger are unpleasant emotions, anger is a dominant emotion, while fear is a submissive emotion.*
>
> *Return a score in the format: {valence, arousal, dominance}, where valence, arousal, and dominance are scalars between -1 and 1. IMPORTANT. STRICTLY STICK TO THE FORMAT AND RETURN IT WITH NO ADDITIONAL CONTEXT OR DECORATIONS.*

Text: `1`"

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